

Cloud for India Region. He cites the example of AlphaFold2 by Google's Deep Mind unit, which leverages AI to predict effective protein shapes for new drugs, to explain how pharma firms can leverage this kind of collaboration.

Vas Narasimhan, Global CEO of Novartis, says this kind of R&D is here to stay. "Clearly, our industry [pharma] is getting reshaped. Tech such as AI and data science are going to impact drug discovery in the long run. These are tools to accelerate drug discovery."

50 | **JUST LIKE PHARMA** firms, healthcare service providers have also taken to tech- and cloud-based solutions in a big way. For instance, Apollo Hospitals Enterprise is using Google Cloud's analytical systems to manage data of over 14 million patients and analyse their health patterns. This helps doctors provide data-driven clinical insights. "We are also working closely with Manipal Hospitals to improve patient care, clinician experiences, and improve the hospital chain's network efficiency. Manipal Hospitals is leveraging Google Cloud to enable virtual care services across its chain of hospitals in the country, creating a new e-pharmacy platform that allows patients to order medicines directly from the hospital, and building remote patient monitoring to improve overall care," says Bedi.

With cloud platforms and tech firms betting big on the Indian healthcare sector, it offers a significant opportunity for both to provide innovative solutions for improved healthcare outcomes, reduced costs for patients and hospitals. For instance, hospital chain Fortis Healthcare is using Microsoft Cloud for Healthcare for internal and service delivery purposes. Tom McGuinness, Cor-

porate Vice President of Global Healthcare & Life Sciences at Microsoft, says the solution provides integrated AI capabilities at scale to make it easier to improve the entire experience.

Then there are the new kids on the block—tech-based start-ups in the medical and pharma space. Chandok of AWS says it is working with several start-ups like Dozee—which provides an AI-based contactless and continuous remote patient monitoring and early warning system—and Practo, which provides services ranging from teleconsultations and booking appointments to lab tests and surgeries.

Even tech giant Apple is eyeing India for its talent pool in research and start-ups. "We are looking for amazing opportunities in healthcare in India and we are really

many as 7,849 health-tech start-ups in India as on January 11, 2023, according to private market data provider Tracxn. A joint report by Redseer Strategy Consultants, Chiratae Ventures and AWS states that the Indian preventive healthcare market is expected to grow from \$93 billion in 2021 to \$197 billion by 2025, at a CAGR of 22 per cent. Comparatively, the curative healthcare segment, which constituted 64 per cent of the total healthcare spends in India in 2021-22, is growing at 15 per cent.

Amit Kumar Verma, CTO of Practo, says the past one year has been a defining moment for the Indian healthcare and healthtech sector. "Start-ups across all stages are looking for cost optimisation solutions. [And] start-ups want to focus on automation and have more managed services so

Cloud computing, AI/ML, big data, and analytics are helping improve service delivery, database management, R&D processes as well as drug discovery

excited," says Sumbul Desai, Vice President of Health at Apple.

But why are giants like Apple and Amazon so gung-ho about healthtech start-ups in India? The answer lies in the rapid rise of non-communicable diseases (NCD) such as cancer, diabetes, stroke and obesity. And, the rise of preventive healthtech start-ups that are working on solutions to tackle them. According to a study by the Indian Council of Medical Research, NCDs' contribution to all deaths in India increased to 61.8 per cent in 2016 from 37.9 per cent in 1990. That has shot up to 70 per cent in 2022, per independent data platform Statista. And to combat this, there were as

that they are able to be more technically efficient," says Kumara Raghavan, Head of Start-ups at AWS India.

Then there's HealthifyMe, which leverages various AWS services, including AI&ML, hosting, search, monitoring, and alerting. "The possibilities are definitely exciting for platform companies like AWS to help enable start-ups across sectors," says Abhijit Khasnis, Vice President of Engineering at HealthifyMe.

For you and me, all this means better healthcare services and medicines at reduced costs. Here's to (tech-powered) good health. **BT**

@neetu_csharma